Chapter 5

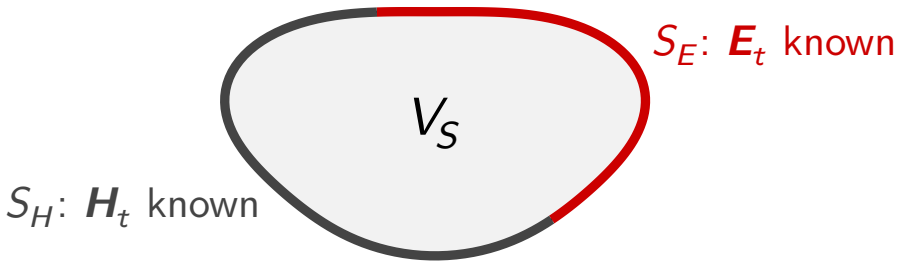
Uniqueness theorem

Uniqueness theorem - statement

Hypotheses: V_S bounded by S; linear isotropic materials, $\varepsilon, \mu, \sigma > 0$ assigned in V_S

Data: we assign

1. **Sources**: $\boldsymbol{E}_i$ is known in V_S for $t > t_0$
2. **Boundary conditions**: for $t > t_0$ we know on S
   - the tangential $\boldsymbol{E}_t$, **or**
   - the tangential $\boldsymbol{H}_t$, **or**
   - $\boldsymbol{E}_t$ on a part S_E of S and $\boldsymbol{H}_t$ on the rest S_H
3. **Initial conditions**: the field is assigned in V_S at $t = t_0$



> **Thesis** (to be proven): the field $(\boldsymbol{E}, \boldsymbol{D}, \boldsymbol{H}, \boldsymbol{B}, \boldsymbol{J})$ is **uniquely** determined for $t > t_0$

Uniqueness theorem - difference fields

By contradiction: suppose two different solutions $(\boldsymbol{E}_1, ..., \boldsymbol{J}_1)$ and $(\boldsymbol{E}_2, ..., \boldsymbol{J}_2)$ of the **same problem**: same sources $\boldsymbol{E}_i$, same boundary and initial conditions

Difference fields: $\boldsymbol{E}_3 = \boldsymbol{E}_1 - \boldsymbol{E}_2, \quad \boldsymbol{D}_3 = \boldsymbol{D}_1 - \boldsymbol{D}_2, \quad ..., \quad \boldsymbol{J}_3 = \boldsymbol{J}_1 - \boldsymbol{J}_2$

Both solutions satisfy Maxwell's equations; subtract them member by member:

$$\nabla \times \boldsymbol{E}_1 = -\frac{\partial \boldsymbol{B}_1}{\partial t}, \quad \nabla \times \boldsymbol{E}_2 = -\frac{\partial \boldsymbol{B}_2}{\partial t} \quad \Rightarrow \quad \nabla \times (\boldsymbol{E}_1 - \boldsymbol{E}_2) = -\frac{\partial (\boldsymbol{B}_1 - \boldsymbol{B}_2)}{\partial t}$$

linearity: $\nabla \times \boldsymbol{E}_1 - \nabla \times \boldsymbol{E}_2 = \nabla \times (\boldsymbol{E}_1 - \boldsymbol{E}_2)$, and the same for $\partial / \partial t$

Same for Ampère-Maxwell: the difference fields satisfy Maxwell's equations

$$\nabla \times \boldsymbol{E}_3 = -\frac{\partial \boldsymbol{B}_3}{\partial t} \qquad \nabla \times \boldsymbol{H}_3 = \boldsymbol{J}_3 + \frac{\partial \boldsymbol{D}_3}{\partial t}$$

Uniqueness theorem - difference fields and sources

Constitutive relations for the difference fields

ε and μ are the same for both solutions:

$$\boldsymbol{D}_3 = \varepsilon \boldsymbol{E}_1 - \varepsilon \boldsymbol{E}_2 = \varepsilon \boldsymbol{E}_3, \quad \boldsymbol{B}_3 = \mu \boldsymbol{H}_1 - \mu \boldsymbol{H}_2 = \mu \boldsymbol{H}_3$$

Both solutions are driven by the **same sources**: the impressed field $\boldsymbol{E}_i$ is a datum of the problem

$$\begin{aligned} \boldsymbol{J}_3 = \boldsymbol{J}_1 - \boldsymbol{J}_2 &= \sigma(\boldsymbol{E}_1 + \boldsymbol{E}_i) - \sigma(\boldsymbol{E}_2 + \boldsymbol{E}_i) \\ &= \sigma(\boldsymbol{E}_1 - \boldsymbol{E}_2) + \cancel{\sigma \boldsymbol{E}_i - \sigma \boldsymbol{E}_i} = \sigma \boldsymbol{E}_3 \end{aligned}$$

> The impressed fields cancel out: **no source** acts on the difference fields, $\boldsymbol{E}_{i,3} = 0$

Uniqueness theorem - Poynting for the difference fields

The difference fields $(\boldsymbol{E}_3, \boldsymbol{D}_3, \boldsymbol{H}_3, \boldsymbol{B}_3, \boldsymbol{J}_3)$ satisfy Maxwell's equations: we can apply the Poynting theorem to them

$$\int_{V_s} \boldsymbol{E}_{i,3} \cdot \boldsymbol{J}_3 \, \mathrm{d}V = \int_{V_s} \frac{J_3^2}{\sigma} \, \mathrm{d}V + \frac{\mathrm{d}}{\mathrm{d}t} \int_{V_s} \left(\frac{B_3^2}{2\mu} + \frac{\varepsilon E_3^2}{2} \right) \mathrm{d}V + \oint_S (\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

No sources, $\boldsymbol{E}_{i,3} = 0$:

$$0 = \int_{V_s} \frac{J_3^2}{\sigma} \, \mathrm{d}V + \frac{\mathrm{d}}{\mathrm{d}t} \int_{V_s} \left(\frac{B_3^2}{2\mu} + \frac{\varepsilon E_3^2}{2} \right) \mathrm{d}V + \oint_S (\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

Both solutions satisfy the **same boundary conditions**, so on S:

- $\boldsymbol{E}_{t,3} = 0$, **or** $\boldsymbol{H}_{t,3} = 0$, **or** $\boldsymbol{E}_{t,3} = 0$ on a part of S and $\boldsymbol{H}_{t,3} = 0$ on the rest

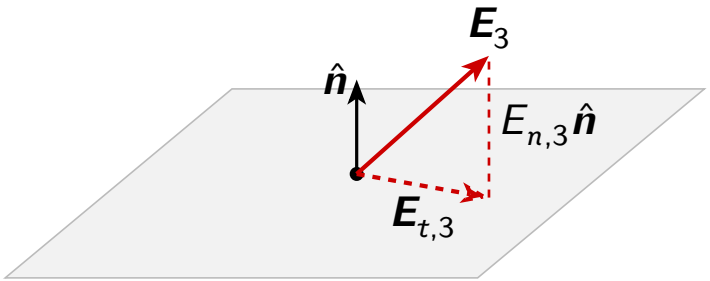
Claim: the **boundary flux** (last term, the flux of the Poynting vector through S) vanishes. Why?

Uniqueness theorem - the boundary flux

Claim: boundary flux $\oint_S (\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}}\, \mathrm{d}S = 0$. Look at the **integrand** at a single point of S.

Split the fields into tangential and normal parts:

$$\boldsymbol{E}_3 = \boldsymbol{E}_{t,3} + E_{n,3}\hat{\boldsymbol{n}}$$
$$\boldsymbol{H}_3 = \boldsymbol{H}_{t,3} + H_{n,3}\hat{\boldsymbol{n}}$$



Substitute into the normal component of the Poynting vector:

$$(\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} = \left[\left(\boldsymbol{E}_{t,3} + E_{n,3}\hat{\boldsymbol{n}}\right) \times \left(\boldsymbol{H}_{t,3} + H_{n,3}\hat{\boldsymbol{n}}\right)\right] \cdot \hat{\boldsymbol{n}}$$

The cross product is **distributive**: one **contribution** for each pair of parts

$$(\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} = \underbrace{\left(\boldsymbol{E}_{t,3} \times \boldsymbol{H}_{t,3}\right) \cdot \hat{\boldsymbol{n}}}_{(1)} + \underbrace{H_{n,3}\left(\boldsymbol{E}_{t,3} \times \hat{\boldsymbol{n}}\right) \cdot \hat{\boldsymbol{n}}}_{(2)} + \underbrace{E_{n,3}\left(\hat{\boldsymbol{n}} \times \boldsymbol{H}_{t,3}\right) \cdot \hat{\boldsymbol{n}}}_{(3)} + \underbrace{E_{n,3}H_{n,3}\left(\hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}}\right) \cdot \hat{\boldsymbol{n}}}_{(4)}$$

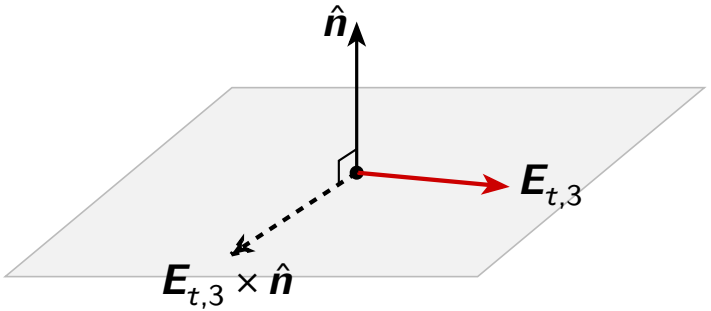
scalar factors ($E_{n,3}$, $H_{n,3}$) can be taken out of the cross product

Uniqueness theorem - contribution (2) vanishes

$$(\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} = \underbrace{(\boldsymbol{E}_{t,3} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(1)} + \underbrace{H_{n,3}(\boldsymbol{E}_{t,3} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(2)} + \underbrace{E_{n,3}(\hat{\boldsymbol{n}} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(3)} + \underbrace{E_{n,3} H_{n,3}(\hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(4)}$$

$\boldsymbol{E}_{t,3}$ lies in the tangent plane, $\hat{\boldsymbol{n}}$ is orthogonal to it (gray plane: tangent plane to S at the point)

A cross product is orthogonal to both factors
$\Rightarrow \boldsymbol{E}_{t,3} \times \hat{\boldsymbol{n}}$ lies **in the tangent plane**



Hence it is orthogonal to $\hat{\boldsymbol{n}}$:

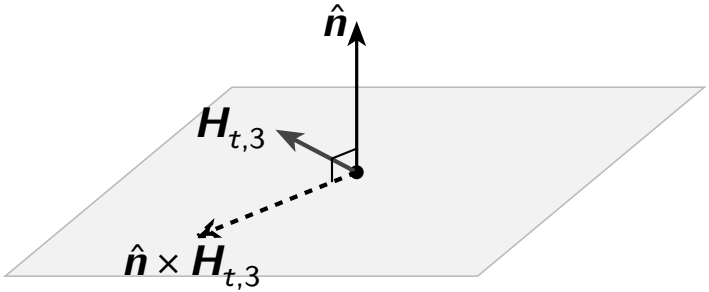
$$(\boldsymbol{E}_{t,3} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}} = 0$$

Uniqueness theorem - contribution (3) vanishes

$$(\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} = \underbrace{(\boldsymbol{E}_{t,3} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(1)} + \underbrace{H_{n,3}(\boldsymbol{E}_{t,3} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(2)} + \underbrace{E_{n,3}(\hat{\boldsymbol{n}} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(3)} + \underbrace{E_{n,3} H_{n,3}(\hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(4)}$$

Same argument, with $\boldsymbol{H}_{t,3}$ in place of $\boldsymbol{E}_{t,3}$

$\hat{\boldsymbol{n}} \times \boldsymbol{H}_{t,3}$ is orthogonal to $\hat{\boldsymbol{n}}$
$\Rightarrow$ it lies **in the tangent plane**



Hence:

$$(\hat{\boldsymbol{n}} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}} = 0$$

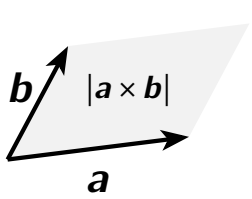
Uniqueness theorem - contribution (4) vanishes

$$(\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} = \underbrace{(\boldsymbol{E}_{t,3} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(1)} + \underbrace{H_{n,3}(\boldsymbol{E}_{t,3} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(2)} + \underbrace{E_{n,3}(\hat{\boldsymbol{n}} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(3)} + \underbrace{E_{n,3}H_{n,3}(\hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(4)}$$

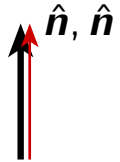
$|\boldsymbol{a} \times \boldsymbol{b}| = |\boldsymbol{a}|\ |\boldsymbol{b}| \sin\theta$ is the **area** of the parallelogram spanned by $\boldsymbol{a}$ and $\boldsymbol{b}$

$\hat{\boldsymbol{n}}$ with itself: $\theta = 0$, the parallelogram collapses

$$\hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}} = \boldsymbol{0}$$



area $= |\boldsymbol{a}||\boldsymbol{b}| \sin\theta$



$\theta = 0 \Rightarrow$ area $= 0$

Uniqueness theorem - contribution (1): tangential fields

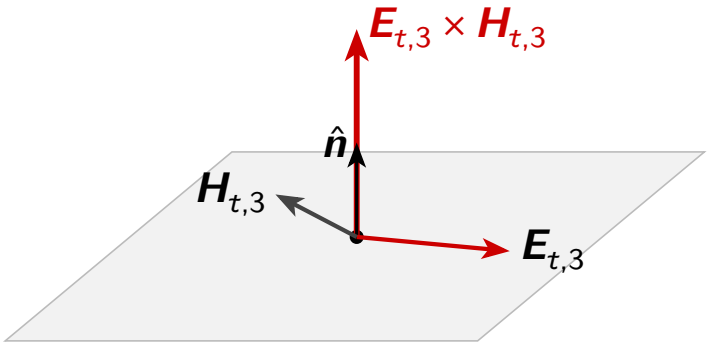
$$(\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} = \underbrace{(\boldsymbol{E}_{t,3} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(1)} + \underbrace{H_{n,3}(\boldsymbol{E}_{t,3} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(2)} + \underbrace{E_{n,3}(\hat{\boldsymbol{n}} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}}}_{(3)} + \underbrace{E_{n,3} H_{n,3}(\hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}}) \cdot \hat{\boldsymbol{n}}}_{(4)}$$

(2), (3), (4) vanish: only contribution (1) is left in the integrand

$\boldsymbol{E}_{t,3}$ and $\boldsymbol{H}_{t,3}$ both lie in the tangent plane
$\Rightarrow$ their cross product is **along** $\hat{\boldsymbol{n}}$

In general this contribution is **not** zero

Only the **tangential** components of $\boldsymbol{E}$ and $\boldsymbol{H}$ carry power through S (this is why tangential fields are the natural boundary data)



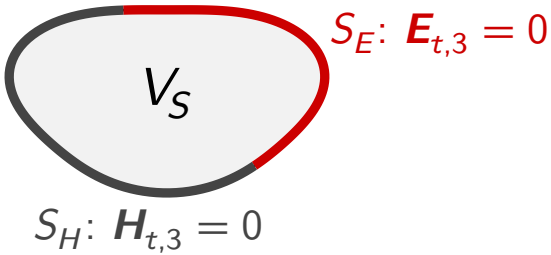
Uniqueness - contribution (1) and boundary conditions

Both solutions satisfy the **same** boundary conditions

On S_E: $\boldsymbol{E}_{t,1} = \boldsymbol{E}_{t,2} \Rightarrow \boldsymbol{E}_{t,3} = \boldsymbol{E}_{t,1} - \boldsymbol{E}_{t,2} = 0$

On S_H: $\boldsymbol{H}_{t,1} = \boldsymbol{H}_{t,2} \Rightarrow \boldsymbol{H}_{t,3} = \boldsymbol{H}_{t,1} - \boldsymbol{H}_{t,2} = 0$

(S_E or S_H may also be the whole S)



At every point of S one of the two factors vanishes:

$$(\boldsymbol{E}_{t,3} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}} = 0 \text{ on } S_E$$
$$(\boldsymbol{E}_{t,3} \times \boldsymbol{H}_{t,3}) \cdot \hat{\boldsymbol{n}} = 0 \text{ on } S_H$$

$$\oint_S (\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S = 0$$

Uniqueness theorem - energy of the difference fields

Energy of the difference fields: $W_3(t) = \int_{V_s} \left(\frac{B_3^2}{2\mu} + \frac{\varepsilon E_3^2}{2} \right) \mathrm{d}V$

Poynting theorem for the difference fields, written with W_3:

$$0 = \int_{V_s} \frac{J_3^2}{\sigma} \mathrm{d}V + \frac{\mathrm{d}W_3}{\mathrm{d}t} + \cancel{\oint_S (\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S}$$

the boundary flux vanishes (previous slides)

Move the Joule term to the left-hand side:

$$\frac{\mathrm{d}W_3}{\mathrm{d}t} = -\int_{V_s} \frac{J_3^2}{\sigma} \mathrm{d}V$$

Uniqueness theorem - energy argument

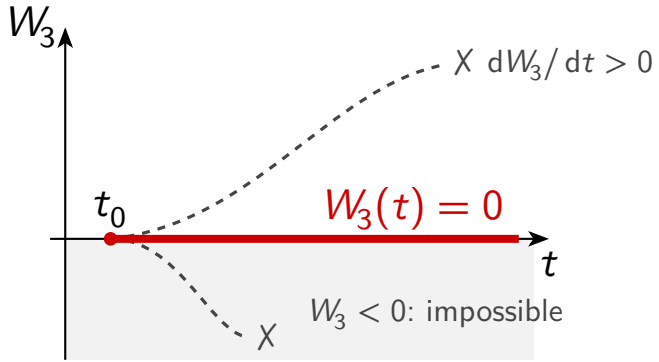
$J_3^2/\sigma \geq 0 \Rightarrow \mathrm{d}W_3/\mathrm{d}t \leq 0$: W_3 can only **decrease**

$B_3^2/(2\mu) \geq 0$ and $\varepsilon E_3^2/2 \geq 0 \Rightarrow W_3(t) \geq 0$

Same initial conditions $\Rightarrow W_3(t_0) = 0$
($\boldsymbol{E}_3 = \boldsymbol{B}_3 = 0$ at $t = t_0$)

Starts from 0, never increases, never negative:

$$W_3(t) = 0 \quad \forall t > t_0$$



Uniqueness theorem - conclusion

$W_3 \equiv 0 \Rightarrow \boldsymbol{E}_3 = 0,\ \boldsymbol{B}_3 = 0$ in V_S

Constitutive relations $\Rightarrow \boldsymbol{D}_3 = \varepsilon \boldsymbol{E}_3 = 0,\ \boldsymbol{H}_3 = \boldsymbol{B}_3/\mu = 0,\ \boldsymbol{J}_3 = \sigma \boldsymbol{E}_3 = 0$

The two solutions coincide: **the solution is unique**

Take-home for modelling: a well-posed e.m. problem needs

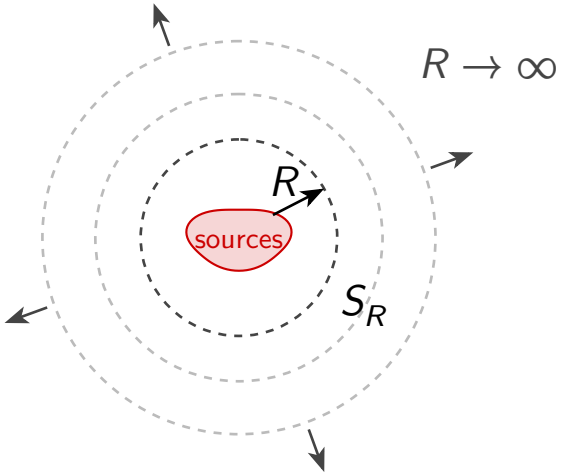
- the **sources** ($\boldsymbol{E}_i$) in the domain
- the **tangential** $\boldsymbol{E}$ or $\boldsymbol{H}$ on every part of the boundary
- the **initial** field (for time-dependent problems)

Uniqueness theorem - unbounded domains

The proof uses a closed boundary S on which the boundary flux vanishes. If $V_S = \mathbb{R}^3$ there is no boundary

Take a sphere S_R of radius R enclosing all the sources and let $R \to \infty$. We have to check that

$$\oint_{S_R} (\boldsymbol{E}_3 \times \boldsymbol{H}_3) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S \to 0 \quad \text{for } R \to \infty$$

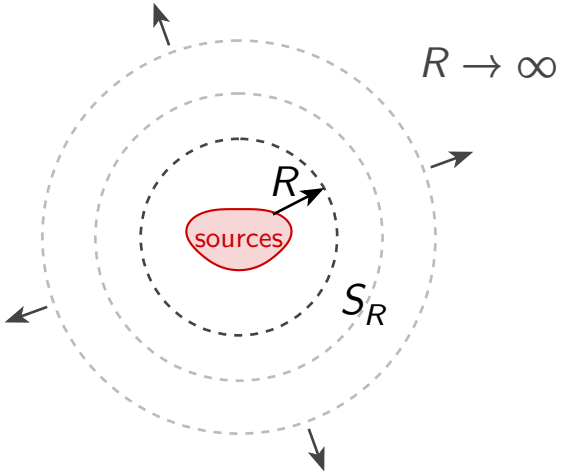


Unbounded domains - area and field decay

How does the boundary flux behave as R grows?

Two opposite effects:

- the area of S_R grows as R^2
- far from the sources the fields decrease



The flux vanishes only if the fields decrease faster than the area grows

next: how fast the fields decrease far from the sources

Unbounded domains - regular conditions at infinity

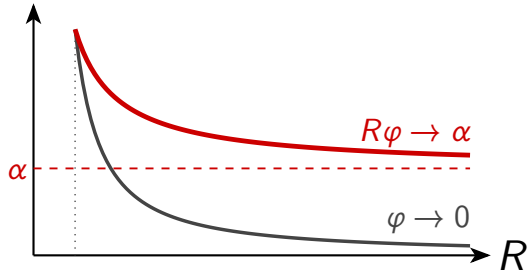
A function φ (the scalar potential, each component of $\boldsymbol{A}$, ...) is **regular at infinity** if, for $R \to \infty$:

$$\lim_{R\to\infty} R\varphi = \alpha, \quad \lim_{R\to\infty} R^2 \frac{\partial \varphi}{\partial r} = \beta, \quad \alpha, \beta \text{ finite}$$

the second limit is the same condition for the derivative, i.e. for the **fields**, which enter the boundary flux

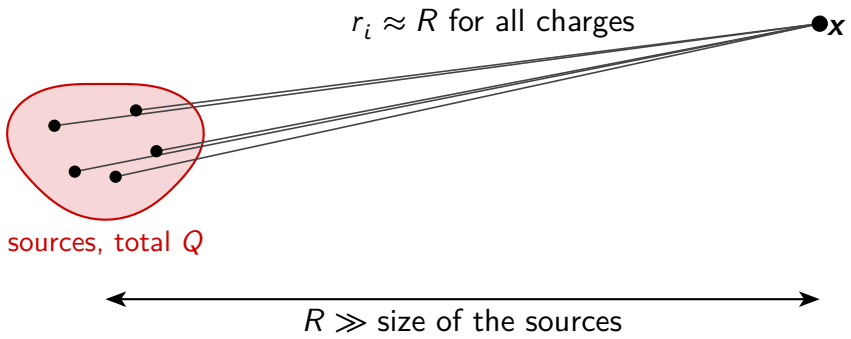
$\varphi \to 0$ is not sufficient, the rate matters. Multiplying by R compares φ with $1/R$:

$\varphi \sim$		$R\varphi \to$
$1/\sqrt{R}$	∞	not regular
$1/R$		finite α
$1/R^2$		0



Unbounded domains - example: electric scalar potential

Example electric scalar potential of a bounded distribution of charges. Very far from it, the distribution looks like a **point**



All charges are at distance $r_i \approx R$ from $\boldsymbol{x}$

Unbounded domains - example: electric scalar potential (cont.)

Far from the sources the potential is that of a point charge Q:

$$\varphi(\boldsymbol{x}) = \frac{1}{4\pi\varepsilon}\sum_i \frac{q_i}{r_i} \approx \frac{1}{4\pi\varepsilon}\frac{\sum_i q_i}{R} = \frac{Q}{4\pi\varepsilon R}$$

Hence φ is regular, with finite limits:

$$R\varphi \to \frac{Q}{4\pi\varepsilon} = \alpha, \qquad R^2\frac{\partial\varphi}{\partial r} \to -\frac{Q}{4\pi\varepsilon} = \beta$$

the same holds for each component of $\boldsymbol{A}$ far from bounded currents; with $Q = 0$ (e.g. a dipole) φ decreases even faster

Unbounded domains - the boundary flux vanishes

Powers of R in the boundary flux through the sphere S_R

quantity	behaviour	why
area of S_R	$\sim R^2$	$4\pi R^2$
$\lvert\boldsymbol{E}\rvert$, $\lvert\boldsymbol{H}\rvert$	$\sim 1/R^2$	derivatives of regular potentials
$\lvert(\boldsymbol{E}\times\boldsymbol{H})\cdot\hat{\boldsymbol{n}}\rvert$	$\sim 1/R^4$	product of the two
boundary flux	$\sim R^2\cdot R^{-4} = 1/R^2$	area × integrand

The boundary flux $\to 0$ for $R\to\infty$: the uniqueness proof holds in the whole space, provided the fields are regular at infinity

Radiating (time-varying) fields decrease only as $1/R$: the flux stays finite ($R^2\cdot R^{-2}$), and **radiation conditions** replace the regularity conditions